Multiplying Two-Digit Numbers

Break both numbers into tens and ones. Four pieces, then add.

Tens times tens

$20 \times 30 = 600$

2 tens × 3 tens: 2 × 3 = 6,
and tens times tens are hundreds.

23 × 14: four pieces

	20	3
10	200	30
4	80	12

10 × 20 = 200
10 × 3 = 30
4 × 20 = 80
4 × 3 = 12

200 + 30 + 80 + 12 = 322

No picture? List the four parts

Tens × tens, tens × ones, ones × tens, ones × ones. Then add.

$31 \times 26 = 600 + 20 + 180 + 6 = 806$

Your turn!

	10	2
10	100	20
3	30	6

12 × 13 = 156

For teachers

Goal: two-digit × two-digit with the area model, then partial products (4.NBT.5). It is packet 56's model with both numbers split. Say “Tens times tens first: those are hundreds.”

Multiplying Two-Digit Numbers: Practice

Split both numbers. Four pieces. Add them.

Part 1 Tens times tens 2 tens × 3 tens = 6 hundreds.

1) $30 \times 40 =$ 1,200	2) $20 \times 70 =$ 1,400	3) $50 \times 60 =$ 3,000
4) $80 \times 30 =$ 2,400	5) $40 \times 90 =$ 3,600	6) $60 \times 20 =$ 1,200
7) $70 \times 70 =$ 4,900	8) $20 \times 50 =$ 1,000	9) $90 \times 30 =$ 2,700
10) $40 \times 40 =$ 1,600	11) $60 \times 80 =$ 4,800	12) $30 \times 50 =$ 1,500

Multiplying Two-Digit Numbers: Practice (continued)

Part 2 The area model The tens and ones are on the outside. Fill in each piece.

13) $24 \times 13 = 312$

	20	4
10	200	40
3	60	12

200 + 40 + 60 + 12 = 312

14) $32 \times 21 = 672$

	30	2
20	600	40
1	30	2

600 + 40 + 30 + 2 = 672

15) $15 \times 42 = 630$

	10	5
40	400	200
2	20	10

400 + 200 + 20 + 10 = 630

Multiplying Two-Digit Numbers: Practice (continued)

Part 2 The area model (continued)

16) $43 \times 26 =$ **1,118**

	40	3
20	800	60
6	240	18

800 + 60 + 240 + 18 = 1,118

Part 3 Split them yourself Write the tens and ones on the outside first.

17) $36 \times 12 =$ **432**

	30	6
10	300	60
2	60	12

300 + 60 + 60 + 12 = 432

18) $27 \times 31 =$ **837**

	20	7
30	600	210
1	20	7

600 + 210 + 20 + 7 = 837

Multiplying Two-Digit Numbers: Practice (continued)

Part 3 Split them yourself (continued)

19) $52 \times 18 = 936$

	50	2
10	500	20
8	400	16

$500 + 20 + 400 + 16 = 936$

20) $41 \times 35 = 1{,}435$

	40	1
30	1,200	30
5	200	5

$1{,}200 + 30 + 200 + 5 = 1{,}435$

Part 4 Write the four parts No picture this time.

21) 34×21

$30 \times 20 = 600$

$4 \times 20 = 80$

$30 \times 1 = 30$

$4 \times 1 = 4$

$600 + 80 + 30 + 4 = 714$

22) 16×45

$10 \times 40 = 400$

$6 \times 40 = 240$

$10 \times 5 = 50$

$6 \times 5 = 30$

$400 + 240 + 50 + 30 = 720$

Multiplying Two-Digit Numbers: Practice (continued)

Part 4 Write the four parts (continued)

23) **53 × 32**

50 × 30 = 1,500
3 × 30 = 90
50 × 2 = 100
3 × 2 = 6
1,500 + 90 + 100 + 6 = 1,696

24) **28 × 17**

20 × 10 = 200
8 × 10 = 80
20 × 7 = 140
8 × 7 = 56
200 + 80 + 140 + 56 = 476

Part 5 Times a tens number No help this time.

25) 34 × 20 = 680
26) 17 × 30 = 510
27) 46 × 40 = 1,840
28) 23 × 60 = 1,380
29) 58 × 20 = 1,160
30) 19 × 50 = 950
31) 62 × 30 = 1,860
32) 35 × 40 = 1,400
33) 71 × 20 = 1,420

Multiplying Two-Digit Numbers: Practice (continued)

Part 6 Two digits times two digits No help this time.

34) $23 \times 15 = 345$	35) $41 \times 12 = 492$	36) $36 \times 24 = 864$
37) $52 \times 13 = 676$	38) $18 \times 27 = 486$	39) $64 \times 11 = 704$
40) $33 \times 32 = 1{,}056$	41) $47 \times 21 = 987$	42) $26 \times 35 = 910$
43) $72 \times 14 = 1{,}008$	44) $15 \times 19 = 285$	45) $38 \times 43 = 1{,}634$

Multiplying Two-Digit Numbers: Practice (continued)

Part 6 Two digits times two digits (continued)

46) $54 \times 26 =$ 1,404

47) $29 \times 31 =$ 899

48) $61 \times 47 =$ 2,867

Part 7 Estimate, then multiply No help this time. Round each number to the nearest ten first.

49) 38×21 is about 800

50) 49×32 is about 1,500

51) 61×19 is about 1,200

52) 27×42 is about 1,200

53) 52×48 is about 2,500

54) 73×29 is about 2,100

Part 8 Multiply

55) $81 \times 16 =$ 1,296

56) $82 \times 36 =$ 2,952

57) $81 \times 44 =$ 3,564

Multiplying Two-Digit Numbers: Practice (continued)

Part 8 Multiply (continued)

58) $32 \times 24 = 768$	59) $24 \times 15 = 360$	60) $58 \times 34 = 1{,}972$
61) $31 \times 33 = 1{,}023$	62) $38 \times 48 = 1{,}824$	63) $76 \times 25 = 1{,}900$
64) $35 \times 19 = 665$	65) $69 \times 26 = 1{,}794$	66) $27 \times 35 = 945$
67) $42 \times 22 = 924$	68) $45 \times 12 = 540$	69) $38 \times 29 = 1{,}102$

Multiplying Two-Digit Numbers: Practice (continued)

Part 8 Multiply (continued)

70) $81 \times 34 =$ 2,754

71) $63 \times 24 =$ 1,512

72) $25 \times 35 =$ 875

73) $58 \times 41 =$ 2,378

74) $28 \times 17 =$ 476

75) $62 \times 35 =$ 2,170

76) $88 \times 23 =$ 2,024

77) $62 \times 27 =$ 1,674

78) $56 \times 13 =$ 728

79) $76 \times 38 =$ 2,888

80) $73 \times 49 =$ 3,577

81) $58 \times 35 =$ 2,030

Multiplying Two-Digit Numbers: Practice (continued)

Part 8 Multiply (continued)

82) $18 \times 18 = 324$

83) $38 \times 16 = 608$

84) $29 \times 15 = 435$

85) $55 \times 22 = 1{,}210$

86) $36 \times 12 = 432$

87) $51 \times 38 = 1{,}938$

88) $75 \times 48 = 3{,}600$

89) $47 \times 31 = 1{,}457$

90) $76 \times 18 = 1{,}368$

Part 9 Remember? Problems from earlier packets.

91) $136 \times 4 = 544$

92) $47 \div 6 = 7$ R 5

Multiplying Two-Digit Numbers: Practice (continued)

Part 9 Remember? (continued)

93) 6,284 to the nearest thousand = 6,000

94) $\frac{3}{4}$ of 24 = 18

95) 3,508 + 2,694 = 6,202

96) $\frac{7}{10}$ > $\frac{3}{10}$

Part 10 Word problems Draw the area model if it helps.

97) A theater has 24 rows with 18 seats in each row. How many seats are there?

Number sentence 24 × 18 = 432

Answer 432 seats

Draw or show your work

98) A baker makes 36 trays of muffins. Each tray holds 12 muffins. How many muffins is that?

Number sentence 36 × 12 = 432

Answer 432 muffins

Draw or show your work

99) A book has 15 chapters of 23 pages each. How many pages does it have?

Number sentence 15 × 23 = 345

Answer 345 pages

Draw or show your work

Multiplying Two-Digit Numbers: Practice (continued)

Part 10 Word problems (continued)

100) Each of 27 kids reads 30 minutes. How many minutes do they read in all?

Number sentence **27 × 30 = 810**

Answer **810** minutes

Draw or show your work

101) Challenge: 12 boxes of 24 pens and 15 boxes of 20 pencils. How many more pencils than pens?

Number sentence **15 × 20 = 300, 12 × 24 = 288, 300 − 288 = 12**

Answer **12** more pencils

Draw or show your work

Check-In: Multiplying Two-Digit Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 The area model

1) $19 \times 22 = 418$

	10	9
20	200	180
2	20	18

$200 + 180 + 20 + 18 = 418$

Part 2 Tens times tens

2) $80 \times 60 = 4{,}800$

3) $50 \times 20 = 1{,}000$

4) $30 \times 70 = 2{,}100$

Part 3 Multiply

5) $47 \times 26 = 1{,}222$

6) $54 \times 36 = 1{,}944$

7) $19 \times 31 = 589$

8) $61 \times 16 = 976$

9) $68 \times 46 = 3{,}128$

10) $33 \times 36 = 1{,}188$